

WASP-12b

R-0172

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_250127 · created 2026-07-16T17:53:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460702.78 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.004 +/- 0.044
Transit depth (Rp/Rs)^2	0.0015 +/- 0.034 [%]
Orbital Inclination (inc)	85.19 +/- 2.87
Airmass coefficient 1 (a1)	0.12 +/- 0.083
Airmass coefficient 2 (a2)	1.63 +/- 0.72
Scatter in the residuals of the lightcurve fit is	69.55 %
Best Comparison Star	None
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.0963 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.004 ± 0.044 vs 0.1178 (deviation 1.98σ)
Duration fit vs published	0.0963 d vs 0.125 d (deviation 6.66σ)
Mid-time O–C	-2.8 min
Depth SNR	0.1
Residual scatter	69.55 %

Light curve

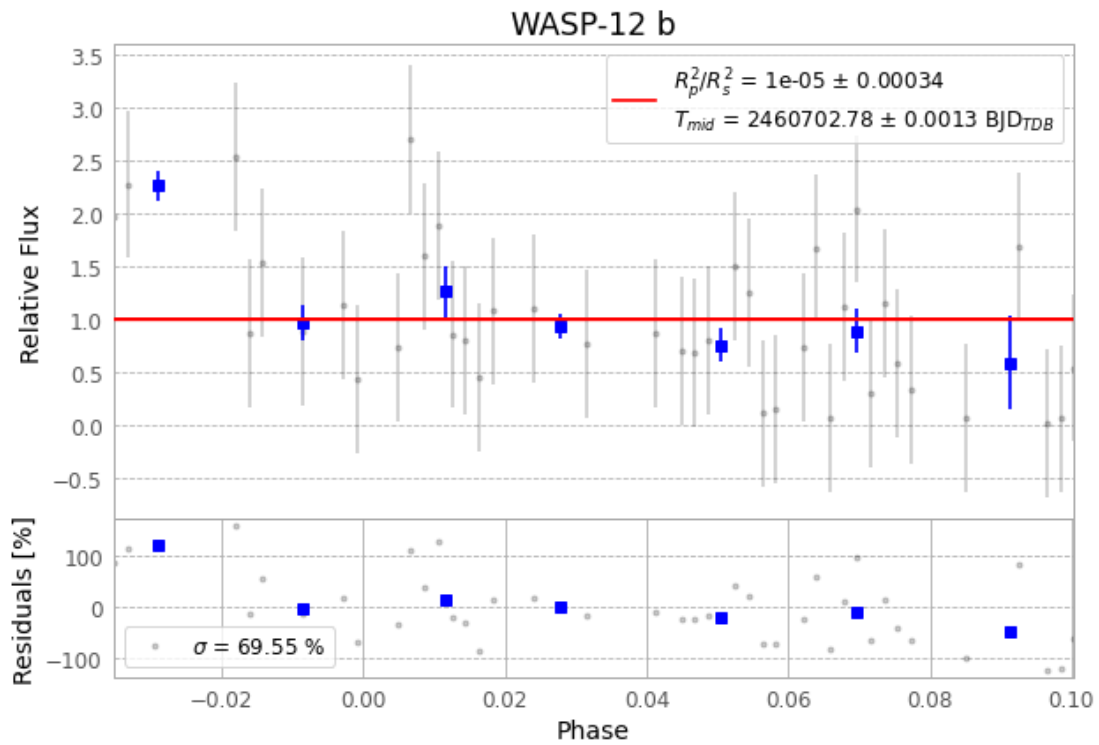


Figure 1 — FinalLightCurve_WASP-12 b_2025-01-27.png (SHA-256 a196ad4a4c6365fe...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 408], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ce6a45a926b151b8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9b8a76a

Data provenance

72 FITS frames (47 MB), WASP-12250127053916.FITS → WASP-12250127091215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-250127.log (18dfe5d97f38...), FinalLightCurve_WASP-12 b_2025-01-27.png (a196ad4a4c63...), FinalLightCurve_WASP-12 b_2025-01-27.csv (f4f980a51808...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:53Z.